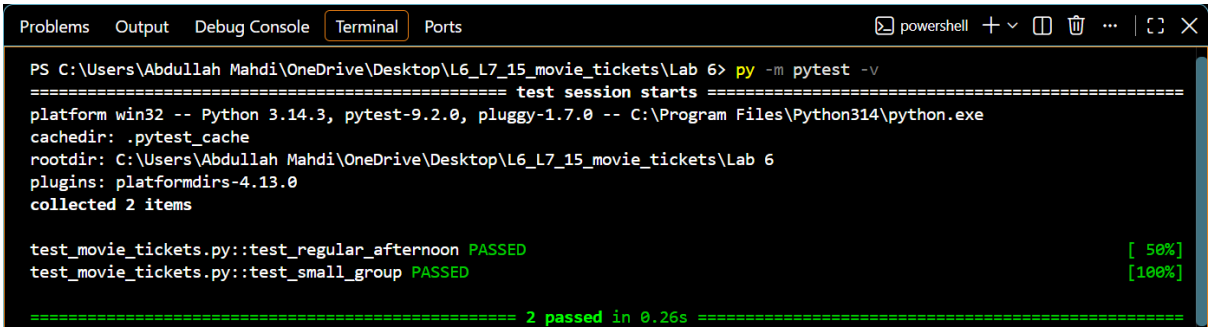
SE LAB - 06

Code and Branch Coverage Analysis

NAME: SHAIKH ABDULLAH MAHDI
SRN: PES2UG24CS452
SECTION: H

GitHub link: https://github.com/AbdullahMahdi29/L6_L7_15_movie_tickets.git

BEFORE:

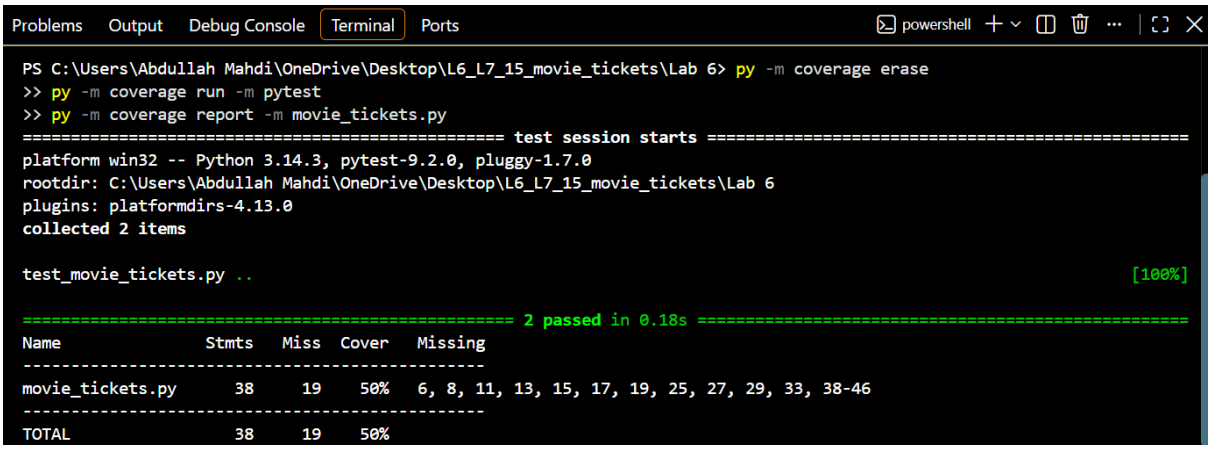


```
Problems  Output  Debug Console  Terminal  Ports                                    powershell

PS C:\Users\Abdullah Mahdi\OneDrive\Desktop\L6_L7_15_movie_tickets\Lab 6> py -m pytest -v
============================= test session starts =============================
platform win32 -- Python 3.14.3, pytest-9.2.0, pluggy-1.7.0 -- C:\Program Files\Python314\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\Abdullah Mahdi\OneDrive\Desktop\L6_L7_15_movie_tickets\Lab 6
plugins: platformdirs-4.13.0
collected 2 items

test_movie_tickets.py::test_regular_afternoon PASSED                     [ 50%]
test_movie_tickets.py::test_small_group PASSED                           [100%]

============================== 2 passed in 0.26s ==============================
```

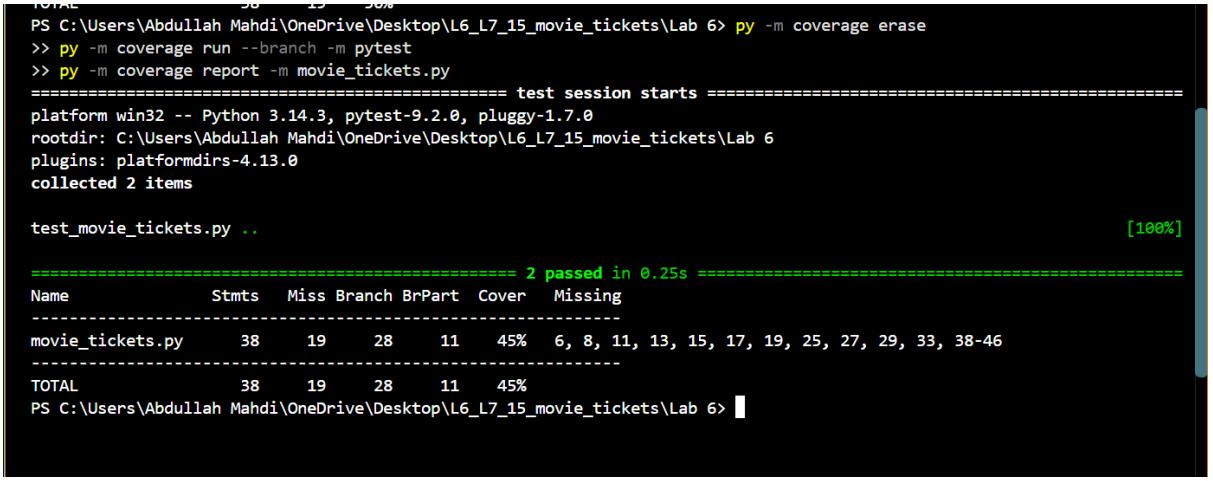


```
Problems  Output  Debug Console  Terminal  Ports                                    powershell

PS C:\Users\Abdullah Mahdi\OneDrive\Desktop\L6_L7_15_movie_tickets\Lab 6> py -m coverage erase
>> py -m coverage run -m pytest
>> py -m coverage report -m movie_tickets.py
============================= test session starts =============================
platform win32 -- Python 3.14.3, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\Abdullah Mahdi\OneDrive\Desktop\L6_L7_15_movie_tickets\Lab 6
plugins: platformdirs-4.13.0
collected 2 items

test_movie_tickets.py ..                                                 [100%]

============================== 2 passed in 0.18s ==============================
Name              Stmts   Miss  Cover   Missing
-----------------------------------------------
movie_tickets.py     38     19    50%   6, 8, 11, 13, 15, 17, 19, 25, 27, 29, 33, 38-46
-----------------------------------------------
TOTAL                38     19    50%
```



```
PS C:\Users\Abdullah Mahdi\OneDrive\Desktop\L6_L7_15_movie_tickets\Lab 6> py -m coverage erase
>> py -m coverage run --branch -m pytest
>> py -m coverage report -m movie_tickets.py
============================= test session starts =============================
platform win32 -- Python 3.14.3, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\Abdullah Mahdi\OneDrive\Desktop\L6_L7_15_movie_tickets\Lab 6
plugins: platformdirs-4.13.0
collected 2 items

test_movie_tickets.py ..                                                 [100%]

============================== 2 passed in 0.25s ==============================
Name              Stmts   Miss Branch BrPart  Cover   Missing
-------------------------------------------------------------
movie_tickets.py     38     19     28     11    45%   6, 8, 11, 13, 15, 17, 19, 25, 27, 29, 33, 38-46
-------------------------------------------------------------
TOTAL                38     19     28     11    45%
PS C:\Users\Abdullah Mahdi\OneDrive\Desktop\L6_L7_15_movie_tickets\Lab 6>
```

AFTER:

code_coverage:

```
PS C:\Users\Abdullah Mahdi\OneDrive\Desktop\L6_L7_15_movie_tickets\Lab 6> py -m coverage erase
PS C:\Users\Abdullah Mahdi\OneDrive\Desktop\L6_L7_15_movie_tickets\Lab 6> py -m coverage run --source=movie_tickets -m pyt
est -v CodeCoverageTest_movie_tickets.py
============================================= test session starts =============================================
platform win32 -- Python 3.14.3, pytest-9.2.0, pluggy-1.7.0 -- C:\Program Files\Python314\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\Abdullah Mahdi\OneDrive\Desktop\L6_L7_15_movie_tickets\Lab 6
plugins: platformdirs-4.13.0
collected 18 items

CodeCoverageTest_movie_tickets.py::test_ticket_price_regular_afternoon PASSED                    [  5%]
CodeCoverageTest_movie_tickets.py::test_ticket_price_morning PASSED                              [ 11%]
CodeCoverageTest_movie_tickets.py::test_ticket_price_evening_weekend PASSED                      [ 16%]
CodeCoverageTest_movie_tickets.py::test_ticket_price_child PASSED                                [ 22%]
CodeCoverageTest_movie_tickets.py::test_ticket_price_young_person_discount PASSED                [ 27%]
CodeCoverageTest_movie_tickets.py::test_ticket_price_senior_discount PASSED                      [ 33%]
CodeCoverageTest_movie_tickets.py::test_invalid_seat PASSED                                      [ 38%]
CodeCoverageTest_movie_tickets.py::test_invalid_show_time_below_zero PASSED                      [ 44%]
CodeCoverageTest_movie_tickets.py::test_invalid_show_time_above_23 PASSED                        [ 50%]
CodeCoverageTest_movie_tickets.py::test_group_discount_no_discount PASSED                        [ 55%]
CodeCoverageTest_movie_tickets.py::test_group_discount_medium_group PASSED                       [ 61%]
CodeCoverageTest_movie_tickets.py::test_group_discount_large_group PASSED                        [ 66%]
CodeCoverageTest_movie_tickets.py::test_group_discount_student_group PASSED                      [ 72%]
CodeCoverageTest_movie_tickets.py::test_group_discount_student_small_group PASSED                [ 77%]
CodeCoverageTest_movie_tickets.py::test_invalid_ticket_count PASSED                              [ 83%]
CodeCoverageTest_movie_tickets.py::test_small_snack_with_coupon PASSED                           [ 88%]
CodeCoverageTest_movie_tickets.py::test_medium_snack_without_coupon PASSED                       [ 94%]
```

```
CodeCoverageTest_movie_tickets.py::test_group_discount_student_small_group PASSED                [ 77%]
CodeCoverageTest_movie_tickets.py::test_invalid_ticket_count PASSED                              [ 83%]
CodeCoverageTest_movie_tickets.py::test_small_snack_with_coupon PASSED                           [ 88%]
CodeCoverageTest_movie_tickets.py::test_medium_snack_without_coupon PASSED                       [ 94%]
CodeCoverageTest_movie_tickets.py::test_large_snack_with_coupon PASSED                           [100%]

============================================== 18 passed in 0.40s ==============================================
PS C:\Users\Abdullah Mahdi\OneDrive\Desktop\L6_L7_15_movie_tickets\Lab 6>
```

```
PS C:\Users\Abdullah Mahdi\OneDrive\Desktop\L6_L7_15_movie_tickets\Lab 6> py -m coverage report -m movie_tickets.py
Name              Stmts   Miss  Cover   Missing
-----------------------------------------------
movie_tickets.py      38      0   100%
-----------------------------------------------
TOTAL                 38      0   100%
PS C:\Users\Abdullah Mahdi\OneDrive\Desktop\L6_L7_15_movie_tickets\Lab 6>
```

branch_coverage:

```
PS C:\Users\Abdullah Mahdi\OneDrive\Desktop\L6_L7_15_movie_tickets\Lab 6> py -m coverage erase
PS C:\Users\Abdullah Mahdi\OneDrive\Desktop\L6_L7_15_movie_tickets\Lab 6> py -m coverage run --branch --source=movie_tick
ts -m pytest -v BranchCoverageTest_movie_tickets.py
============================================= test session starts =============================================
platform win32 -- Python 3.14.3, pytest-9.2.0, pluggy-1.7.0 -- C:\Program Files\Python314\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\Abdullah Mahdi\OneDrive\Desktop\L6_L7_15_movie_tickets\Lab 6
plugins: platformdirs-4.13.0
collected 23 items

BranchCoverageTest_movie_tickets.py::test_morning_price PASSED                                          [  4%]
BranchCoverageTest_movie_tickets.py::test_afternoon_price PASSED                                        [  8%]
BranchCoverageTest_movie_tickets.py::test_evening_price PASSED                                          [ 13%]
BranchCoverageTest_movie_tickets.py::test_weekend_surcharge PASSED                                      [ 17%]
BranchCoverageTest_movie_tickets.py::test_unknown_seat[vip] PASSED                                      [ 21%]
BranchCoverageTest_movie_tickets.py::test_unknown_seat[economy] PASSED                                  [ 26%]
BranchCoverageTest_movie_tickets.py::test_invalid_hours[-1] PASSED                                      [ 30%]
BranchCoverageTest_movie_tickets.py::test_invalid_hours[24] PASSED                                      [ 34%]
BranchCoverageTest_movie_tickets.py::test_child_free_ticket PASSED                                      [ 39%]
BranchCoverageTest_movie_tickets.py::test_child_discount PASSED                                         [ 43%]
BranchCoverageTest_movie_tickets.py::test_adult_full_price PASSED                                       [ 47%]
BranchCoverageTest_movie_tickets.py::test_senior_discount PASSED                                        [ 52%]
BranchCoverageTest_movie_tickets.py::test_zero_tickets PASSED                                           [ 56%]
BranchCoverageTest_movie_tickets.py::test_small_group PASSED                                            [ 60%]
BranchCoverageTest_movie_tickets.py::test_medium_group PASSED                                           [ 65%]
BranchCoverageTest_movie_tickets.py::test_large_group PASSED                                            [ 69%]
BranchCoverageTest_movie_tickets.py::test_student_medium_group PASSED                                   [ 73%]
BranchCoverageTest_movie_tickets.py::test_student_large_group PASSED                                    [ 78%]
BranchCoverageTest_movie_tickets.py::test_student_small_group PASSED                                    [ 82%]
BranchCoverageTest_movie_tickets.py::test_small_snack PASSED                                            [ 86%]
```

```
BranchCoverageTest_movie_tickets.py::test_student_medium_group PASSED                                   [ 73%]
BranchCoverageTest_movie_tickets.py::test_student_large_group PASSED                                    [ 78%]
BranchCoverageTest_movie_tickets.py::test_student_small_group PASSED                                    [ 82%]
BranchCoverageTest_movie_tickets.py::test_small_snack PASSED                                            [ 86%]
BranchCoverageTest_movie_tickets.py::test_medium_snack_with_coupon PASSED                               [ 91%]
BranchCoverageTest_movie_tickets.py::test_large_snack_without_coupon PASSED                             [ 95%]
BranchCoverageTest_movie_tickets.py::test_unknown_snack_size PASSED                                     [100%]

============================================== 23 passed in 0.39s ==============================================
```

```
PS C:\Users\Abdullah Mahdi\OneDrive\Desktop\L6_L7_15_movie_tickets\Lab 6> py -m coverage report -m movie_tickets.py
Name               Stmts   Miss Branch BrPart  Cover   Missing
--------------------------------------------------------------
movie_tickets.py      38      0     28      0   100%
--------------------------------------------------------------
TOTAL                 38      0     28      0   100%
PS C:\Users\Abdullah Mahdi\OneDrive\Desktop\L6_L7_15_movie_tickets\Lab 6>
```